

Velocity Vortex
Engineering Folio
National Final 2026

velocity vortex



Re-Engineering Australia
Foundation



John Gardner MP
MEMBER FOR MORIALTA



BocaBearings



Engineering Philosophy

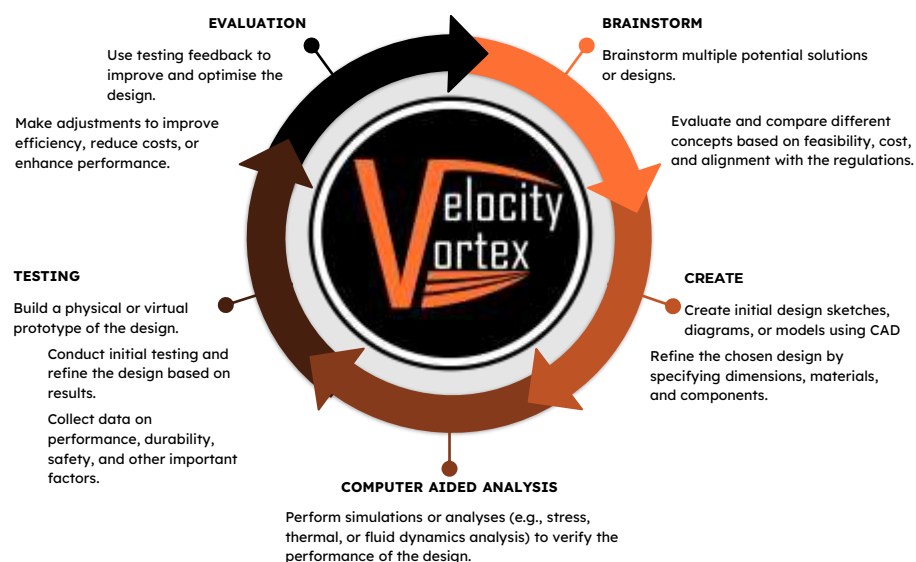
Our main objective in this competition is to have designed and manufactured a competitive and compliant race car.. Our objectives include planned, rehearsed and consistent assembly, using marking and measuring tools and jigs to promote quality control and accuracy. We will strive to achieve our car target mass, and ultimately to achieve consistent and competitive race times. We will use CAD and third party software to generate our CAD designs and to test and analyse them. We have used the basics of motion, and an understanding of the physical losses with STEM Racing cars.



Engineering Cycle

Our engineering cycle was at the centre of our creative car design process. We used the model below on all aspects of the engineering process including individual components and concepts on materials and manufacturing processes.

This allowed us to produce the most efficient aerodynamic car we possibly could whilst adhering to the mass requirements and compliance regulation of the competition.



Newton's Laws of Motion

Inertia - First Law

"An object at rest or in motion remains so unless acted upon by an external force" our F1 in schools car is initially at rest. The CO2 canister then launches the car, acting as an unbalanced external force, overcoming the car and producing inertia. (Glenn Research Center, 2024) NASA

Force - Second Law

"Acceleration is directly proportional to force and inversely proportional to mass ($a=F/m$).\" the car's acceleration is depend on the force it experiences and its mass. The cars force is the CO2 canister which we cannot control, although the mass of the car we can control. Therefore by having the lightest possible STEM Racing car we increase our car's acceleration. (, Glenn Research Center, 2024) NASA

Action and Reaction - Third Law

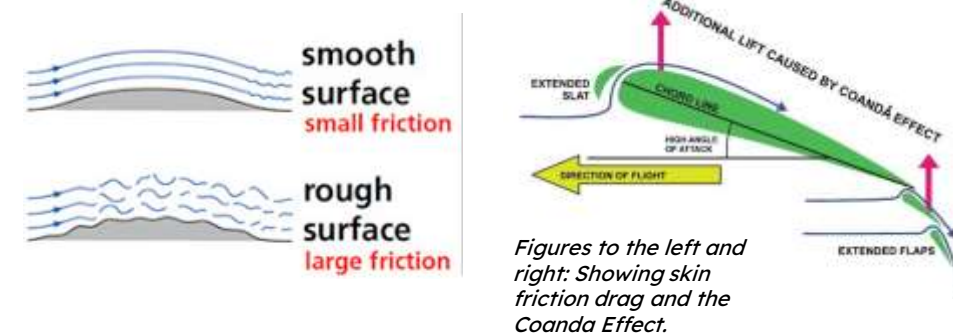
"For every action, there is an equal and opposite reaction." When the CO2 is released, it propels the car forward with an equal and opposite force pushing it down the track. This reinforces the importance of harnessing the available energy from the canister (, Glenn Research Center, 2024) NASA

Profile Drag

Profile, or pressure drag, is primarily influenced by an object's shape and size, and is proportional to the pressure difference between the front and rear surfaces, including the frontal area. It occurs when airflow separates abruptly from the body, creating pressure imbalances and resulting in induced drag. Understanding this phenomenon helps us design for smooth, laminar flow and minimise disturbances in the airflow. (Nuclear-Power, 2023)

Skin Friction Drag

Skin friction drag is the resistance caused by friction between a fluid and the surface of a moving object. It is directly related to the surface area in contact with the fluid and increases with the square of the object's velocity. Our finishing system is designed to reduce this effect as much as possible. (SKYbrary, 2024)

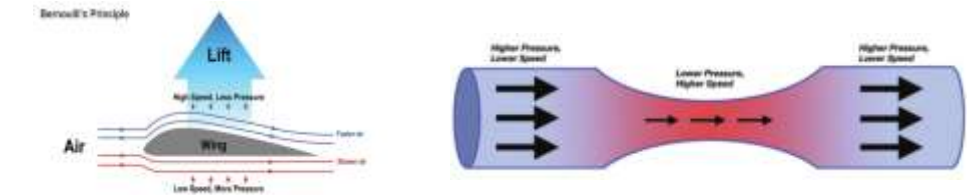


Coanda Effect

The Coanda Effect occurs when a jet of fluid attaches itself to a nearby surface and continues to follow it, even as the surface curves away. This phenomenon is important to understand when designing our car, as we aim to promote laminar flow around the body and minimise areas that could generate turbulence. Being aware of the Coandă Effect is especially crucial when shaping the rear of the car, to prevent unwanted or unpredictable airflow behaviour. (S & C Thermofluids, 2023)

Bernoulli's Principle

When fluid flows through a restricted space, such as around a wing or wheel assembly, it accelerates, creating a pressure difference that increases drag. Since the total energy of the fluid remains constant, we aim to minimise these pressure differentials to reduce drag. (Byjus, 2024)



Reynolds Number

Reynolds number represents the ratio of inertial to viscous forces within a fluid and is used to indicate the laminar or turbulent nature of a flow. It is represented by:

$$Re=(v*I)nu.$$

We found that in isolation, Reynolds number was not that meaningful when studying flow around an object, mainly because STEM cars don't reach meaningful speeds for this analysis.

Laminar Flow

Laminar flow occurs when a fluid moves smoothly around an object in parallel layers with minimal disruption. In contrast, disturbed flow causes turbulence, pressure differences, and increased drag. Our car designs take these flow types into account, and we have worked to minimise aerodynamic losses wherever possible



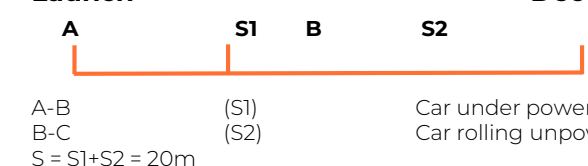
Tipping Moment

The tipping moment occurs at launch when the canister force acts above the front axle, causing the car to pivot forward since the thrust line and centre of mass are misaligned. This can reduce stability, increase friction and drag, overload the front wheels, and momentarily disrupt aerodynamics. To minimise tipping, we aim to position the centre of mass rearward and align it closely with the canister's thrust line.

Acceleration

Through extensive track testing, we calculated that our car remains in the acceleration phase for the first 5-7 metres using standard 28.5g canisters. Maximising this phase is crucial, as the car must reach peak acceleration before the pressure in the canister reaches equilibrium with the atmosphere.

Launch



Figures above: Showing race summary.
Credit Peter Brown.

Deceleration

The Canister

$$F_G = mG aG$$

Figure above: Shows the summation of the race and launch event. Credit ACRUX.

When the rear of the cylinder is punctured, the pressured body of gas, (the mass of gas) accelerating out of the hole creates a rearward force F_G . The pressurised gas release creates a reaction back on itself resulting in the encapsulated canister producing a forward force on the car body. Newton's 3rd Law. Average mass for a world cylinder contains 8 grams charge of compressed air and weighs between 28.9g and 29.4g T1.20 (Ref: Peter Brown)

Wheel Alignment

Wheel alignment plays a critical role in keeping a car tracking straight, allowing our car to race to its potential. Our car must ensure that the wheels are correctly aligned helping to reduce body yaw—the unwanted rotation of the car around its vertical axis—improving directional stability and overall performance. Proper alignment not only contributes to faster lap times but also ensures compliance with competition regulations.

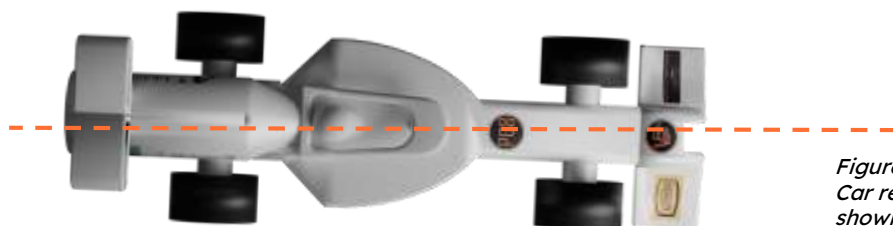


Figure to the left: Car render showing yaw from Blender.

Target Mass

Velocity Vortex takes a systematic approach to achieving the target mass in our car design, aiming for 50.5 g (± 0.2 g) to optimise acceleration. Using the equation $a = F/m$ (where a = acceleration, F = force from the canister, and m = car mass), we understand the direct impact of mass on performance. Our school track testing shows that each additional gram adds roughly 0.003 seconds to lap time, so minimising mass is key to improving speed. Even small differences can have a significant effect, as force is determined by the change in momentum ($F = m \times \Delta v$), with mass and acceleration being critical factors.

Computational Fluid Dynamics

We employed Computational Fluid Dynamics (CFD) to design and analyse the aerodynamics of our car. Using software such as Ansys, Autodesk CFD, and Virtual Wind Tunnel, we obtained detailed and rigorous data that guided and informed the evolution of our design concepts.

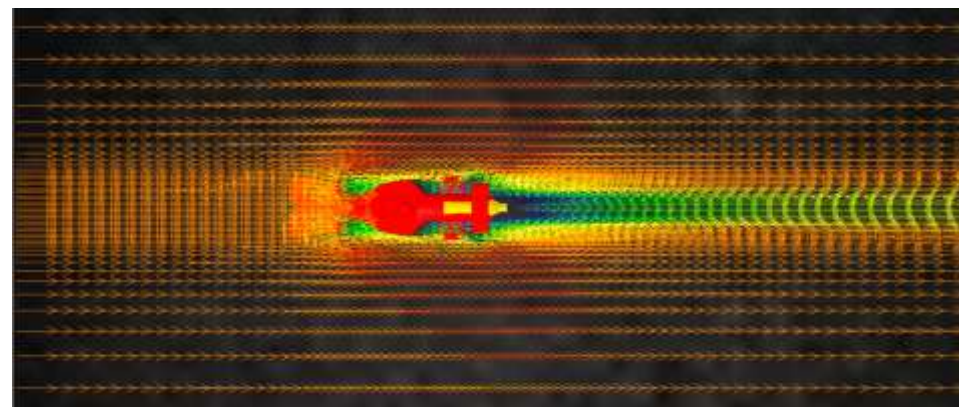


Figure above: Image from VWT of the final car.

Principles of Aerodynamics

By definition, the study of aerodynamics focuss on the shapes of objects that move through the air. We will use the range of software previously mentioned to gather data to help design components, body shape and acheive overall car design. Data we use includes, downforce, lift, frontal area and drag coefficient, which is a summary of all forces acting on the car. It is this data (C_d) that we mainly use. Although we know that from testing our class cars mass and wheel and axle performance are bigger influences on final race times than streamlined performance, we have designed our cars around sound aerodynamic principles, understanding profile drag in particular.

Finite Element Analysis

Finite Element Analysis (FEA) is a computer-based simulation used to predict how a product will respond to real-world forces, vibrations, heat, fluid flow, and other physical effects. We applied FEA to the halo, front and rear wing assemblies, wheel and axle system, and tether guides. (Autodesk, 2024). These tests were mainly conducted to confirm what our track testing indicated, but also provided evidence of digital analysis, using Fusion CAD software.

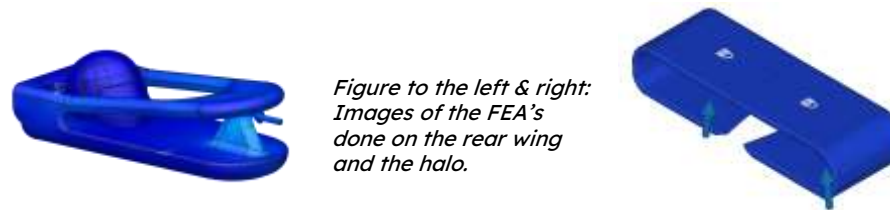


Figure to the left & right: Images of the FEA's done on the rear wing and the halo.

Start Alignment Jigs

Accurate and true alignment of the car on launch is critical. Having the thrust driving through the X axis on the car when positioned on the track will minimise tether and other losses and will maximise the acceleration phase. We have created jigs to assist us with this known loss.

Power supply

The power supply is a Carbon Dioxide cylinder weighing approximately 28gms, containing approximately 8gms of gas under pressure. They vary in gas volume and sometimes in manufacture and are therefore not a reliable source of energy. STEM Racing quality control our race cylinders by weighing before competition.

Application of 3D printing

The use of 3D printing in STEM racing is a crucial part for our manufacturing and prototyping design processes. We have good access to slicing software and a lab of 3D printers. In addition we have a Resin printer that we use for added accuracy and minimal distortion. We have experimented with deposition rates and different filaments, but our results have been very reliable and accurate using default settings like positioning on the table and Fill %'s.

Regulations

From examining the rules and regulations and taking advice from other, more experienced teams from our school, we have understood the importance of staying within the regulations for our car to avoid points loss and racing time penalties. Initial checks were always made using the Drawing workspace in Fusion, but after prototype manufacture, we continuously checked using digital verniers and height gauges, as well as our jigs and drill bits etc. We found these after manufacturing checks are the most reliable.

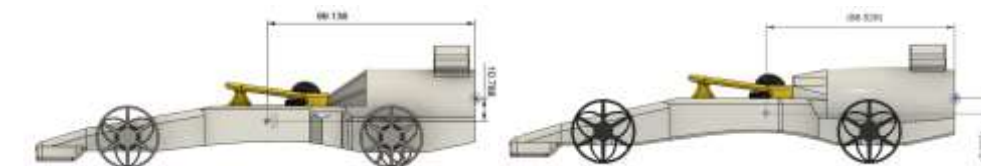
Tether Guides

We also looked into various different materials to test their dynamic friction coefficients but our existing Brass eyelets performed well at the States, so we stayed with them. We noted that the tether guides should positioned be as far apart as possible for the best stability while also not too low to fail T3.8 and be in contact with the track.

Centre of Mass

The CAD model gives a digital centre of mass (CoM) position along all axes, but checking the actual machined and assembled cars is important, as we aim to get the CoM reward and towards the x axis vector to minimise the tipping moment. (New World Encyclopedia, 2024)

Figures below: Images from Fusion 360 showing the CoM dimensions front the origin on our state and national cars.



Rotational Inertia

The moment of inertia or more widely known as rotational inertia is the ratio between the torque and the resultant acceleration around the axis of in our case, our wheels. All bodies in motion are subjected to the properties of rotational inertia. To help, we have designed the wheel so that the mass is centred around the hub, whilst the outer rim thickness is kept to a minimum. Also, the axles don't rotate.

Forces Acting on the Car

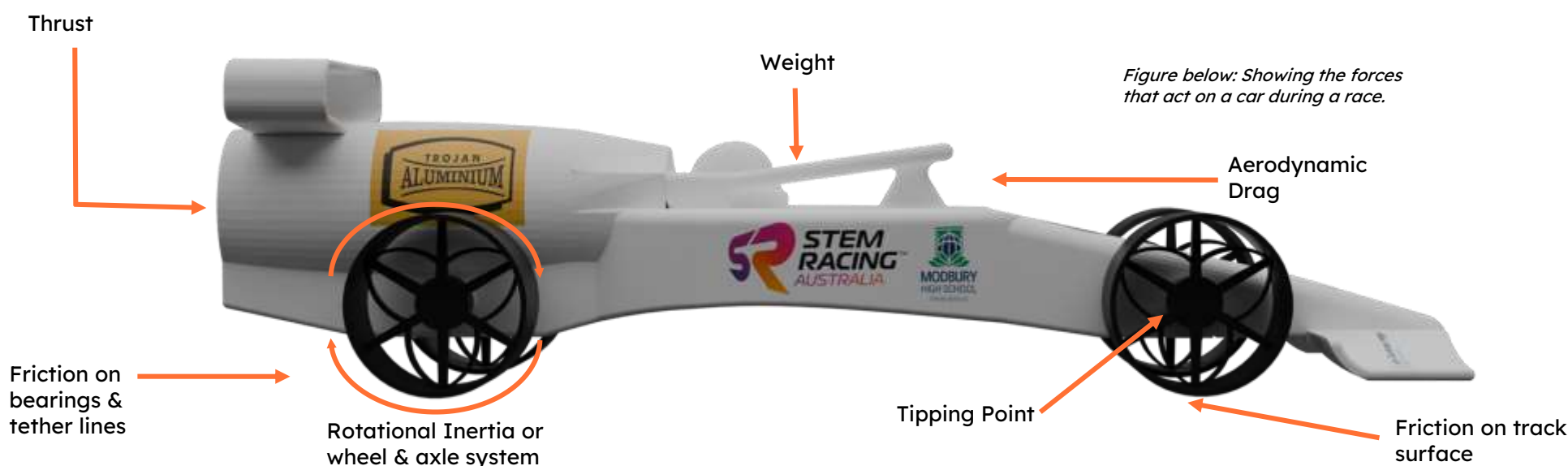


Figure below: Showing the forces that act on a car during a race.

While designing our car we went through multiple design changes for our front wings, back wings, core body and side pods, as seen below. We have outlined the different designs which were used as the car's design evolved.

Front Wings

Concept 1

The first concept for our front wing originated from Liam's car from his F1 in schools class. This was used as a basic template to expand to a more effective, lighter and intricate wing design. Our main problem with this wing was its mass. We also felt that we could improve the aerodynamics. This concept is compliant.

Figure to the right: Showing a front wing prototype from fusion 360.



Concept 2

This front wing incorporated a multi vein design, and we liked this design as it mirrored a real F1 wing look, but it was still too heavy.. We wanted to design a wing which incorporated the best parts of these two but with mass reduction. This concept was compliant but its mass also compromised the tipping point to far forward.



Figure to the left: Showing a front wing prototype from fusion 360.

State & Nationals Front Wing

This is our state finals wing which we are reusing for nationals. It is a redesign of our first prototype blending a simple design but with reduced mass. The aerodynamic features such as its arched wings designed for guiding the air to flow either up and over or around the front set of wheels is a proven positive feature. As the wing is one solid piece unlike our second prototype with multiple sections, it is stronger and less likely to fail during deceleration.

Figure to the right: Showing the final front wing prototype from fusion 360.

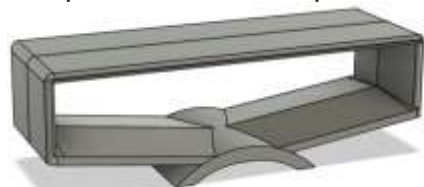


Rear Wing

Concept 1

Our first concept design for a back wing is quite simple with the main focus being to create an attachment point for the wing to slot into the back canister housing of the car. The design was used as a prototype and was compliant. This wing is has too much mass and not was not aerodynamic which we aimed to improve upon in future concepts

Figure to the right Showing a rear wing prototype from fusion 360.



State Final Back Wing

This is our state final back wing where we have rounded/filleted everything to improve aerodynamic performance and have reduced wall thickness, but still met the regulations. It was aesthetically pleasing, with its rounded edges and minimal surface area. These combined modifications resulted in a wing that also met our mass target.

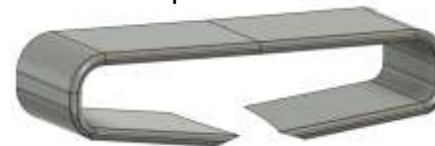


Figure to the left: Showing a rear wing prototype from fusion 360.

National Final Back Wing

Our national final back wing is fundamentally the same wing was the state finals with one major difference. We have removed the attachment point from the wing because of the cars new rounded canister housing. Meaning the only way to sit the wing level on the canister housing was to remove the attachment point

Figure to the right: Showing the final rear wing prototype from fusion 360.



Side pods

Concept 1

The first concept design had the intended purpose of this design to allow air to flow along the floor and around the side pods giving multiple paths for the airflow to take. Though this design was only a concept and was more of a base design to expand off of for a more effective design, it was compliant.

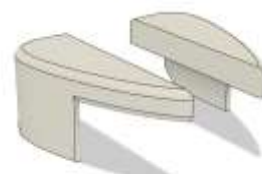


Figure to the left: Showing a side pod prototype from fusion 360

Concept 2

In our second concept we decided to step away from the open air space underneath the side pods because it was very difficult to fit in the virtual cargo. We chose to close the space on the floor of the side pods in our next design. The side pods were then designed to have a rear straight edge then curving around its sides around to the front. This was an aerodynamic feature that also reduced mass. Concept #1 had a very tight curve at the end and we felt cutting it straight would give a more generous path for the airflow. It is compliant.

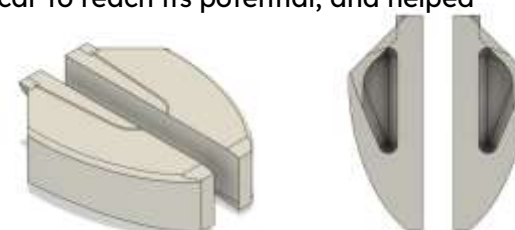
Figure to the right: Showing a side pod prototype from fusion 360



State Final Side Pod

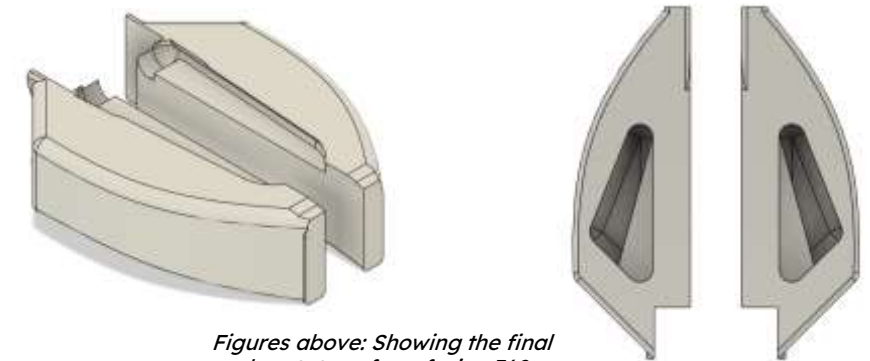
Our state final side pod design was changed for to two major reasons. It's lightweight and it's aerodynamic capabilities combined with the car to produce a fast lightweight car. Simultaneously, underneath the wing the curvature aligns with that of the car. Hence, choosing this design was a crucial step as it allowed car to reach its potential, and helped with overall mass restrictions.

Figures to the right: Showing our state final side pods from fusion 360



National Side Pods

Our national side pods, similar to our national wing, is fundamentally the same to our state finals version. As we created a whole new body for nationals, the side pods were remade into the new car but still with all the benefits the state finals variant provided.



Figures above: Showing the final pod prototype from fusion 360

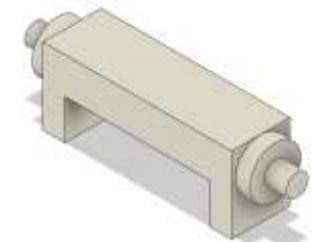
Axle System

At the state finals we used carbon fiber rods glued to the inside of the car body as an axle system. For the National final, our new system will be based on a stub axle rather than the common axle we used previously. The change was primarily for mass target reasons.

Prototype 1

The first prototype is quite simple in design and function. With two slots on each side for bearings / wheels to be slid into, and an arch type structure connecting the two segments.

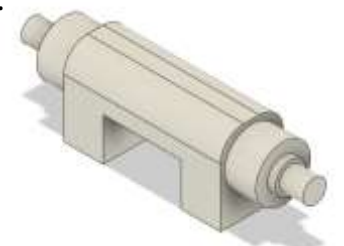
Figure to the right: showing our axle system prototype from fusion 360.



Final Axle system

We made modifications to the final axle system. A one mm stopper has been placed around the bearing housing to reduce friction when the bearing spins. The top has also been curved due to be housed neatly in the car body, which will be machined using a 6mm ball mill. This will enable a more reliable, accurate assembly.

Figure to the right: showing our final axle system from fusion 360.



Final Axle System- extended

An issue needing fixing was that our axle system wasn't long enough for the back of the car. This would have resulted in the wheels making contact with the canister housing. Our solution was to use the same axle system but to lengthen the component to provide clearance.

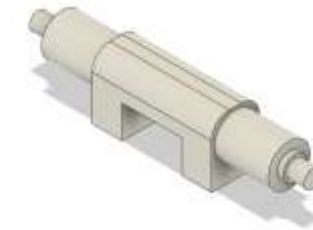


Figure to the left: showing our final axle system with extended axles from fusion 360.

Car Body philosophy

Understanding the link between mass and velocity, we designed our car bodies to stay within the ideal 38–42 g range, with 70 - 85gm blocks. Allowing the final car to reach our 50g target after assembly and the finishing processes. Machinability was checked using the Accessibility and Minimum radius inspection tools, using our 6mm Ball Mill. Aerodynamics was also a factor but we accepted a reasonable Cd as a compromise for the above factors. Our final car focused on a compact body similar to our first prototype, providing a smaller frontal area which encourages laminar flow and to naturally guide air around the car. We also chose to remove rear side pods to reduce drag behind the rear wheels and minimise overall mass. This compromised the tipping moment however.

State Finals Car Body

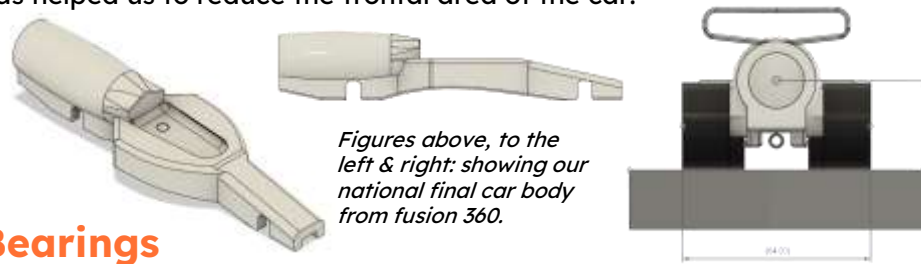
The state finals car featured the addition of a funnel-shaped section behind the halo to eliminate the flat face of the canister housing and to guide airflow smoothly around it. Another was thinning the undercarriage/floor on the final body to further optimise performance, aerodynamics and mass.



Figures above, to the right & left: showing our state final car body from fusion 360

National Finals Car Body

Our nationals finals car body is a direct successor to our state finals car. While the nationals car largely reuses wings and wheels from states, the body has been completely redesigned. The nationals car is an evolution of our state car, where we learnt much about compliant, aerodynamic and mass target designing. Key improvements include our lofting behind the halo. We also lowered the canister by **2.3mm** and added a gentle camber which helped with reduced frontal area and a better laminar flow of air. Other changes featured on the car body included changes to the floor/underside of the car to house the Virtual Cargo, and improve aerodynamics by filleting all sharp edges. Due to our new axle system we were able to bring the width of the wheelbase in by **10mm** which again has helped us to reduce the frontal area of the car.

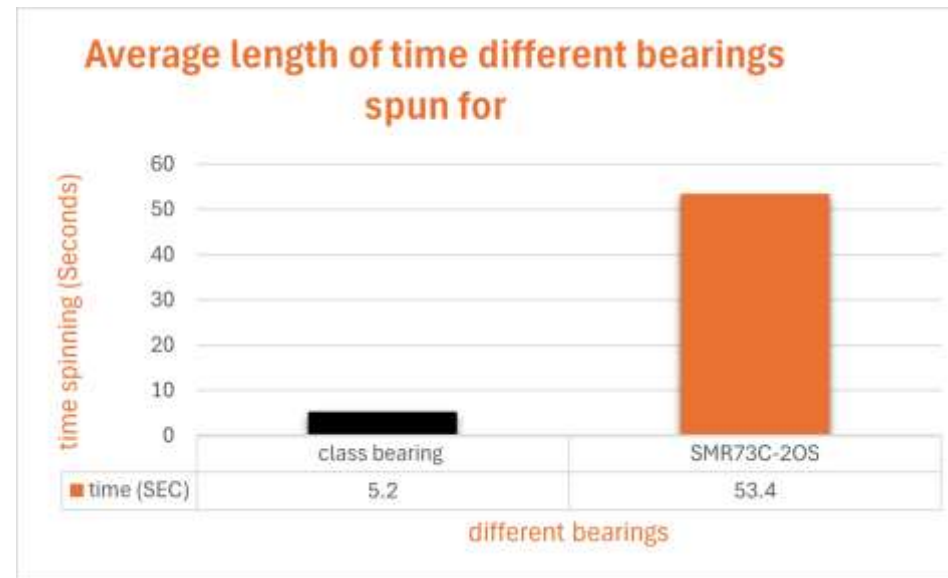


Figures above, to the left & right: showing our national final car body from fusion 360.

Bearings

We purchased our bearings from **Boca: SMR73C-20S #7 LD (3x7x3)**. Hybrid ceramic bearings use steel rings with ceramic balls, offering lower friction and better performance than the all steel bearings.

We chose hybrid ceramics because they provided excellent performance without the high cost of full ceramic bearings—the price difference was over \$40, with minimal expected performance gain. After comparing our class steel bearings with a ceramic bearing in multiple spin-time tests (averaged in the graph below), the hybrid ceramics showed significantly better results, making them the clear choice.



Full Ceramic Steel Class Hybrid

Figure above: Graph showing bearing spin times

Figures to the left: Showing different bearing types

Wheel Designs

Prototype 1

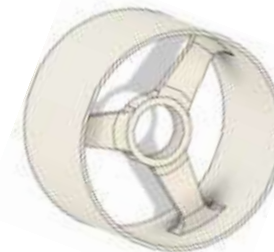
Based in rotational inertia principles, our first wheel design focused on simplicity to achieve a lightweight, high-performing wheel, which featured five supports, providing strength and durability, it was however heavy with too much mass not centrally located



Figures above & below: Showing wheel prototypes from fusion 360

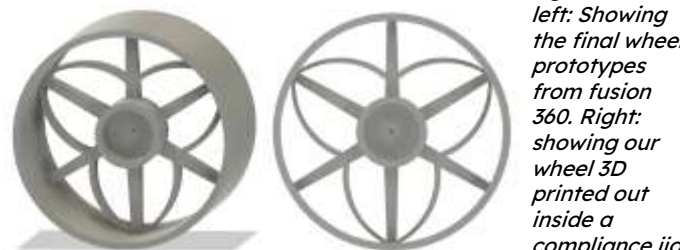
Prototype 2

Our second wheel design aimed to concentrate mass in the hub. We reduced the supports from five to three, increasing their thickness to balance strength and weight. However, testing showed wheel was still too heavy, leading us to attempt to further remove mass from the design.



Final Design

Our final wheel design combined simplicity, classic design, and innovation. It featured six thin, symmetrically placed supports for balance, with curved edges to improve aerodynamics and create a visually appealing design. Despite being thin, the supports' total weight nearly matched our second concept, providing a lightweight, balanced wheel that embodied our team's focus on performance and innovation. This maximized mass around the hub.



Figures to the left: Showing the final wheel prototypes from fusion 360. Right: showing our wheel 3D printed out inside a compliance jig.



Application of CAD

CAD allows us to turn ideas into accurate, testable designs. Without effective use of CAD, STEM Racing cars would struggle with compliance and efficiency. Using Autodesk Fusion, we can model, refine, and virtually test nearly any component, ensuring reliable manufacturing.

Successful CAD work also relies on good organisation, from managing prototype files to tracking changes in sketches and bodies. We prioritise this to streamline our design process. Fusion was used to create all custom car parts, position components like the tether guide, bearings, halo, and helmet, and produce the drawings for our compliance portfolio.

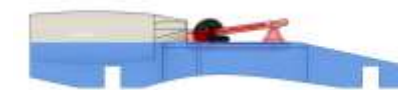


Figure above: Showing the use of the combine and cut tool in fusion 360

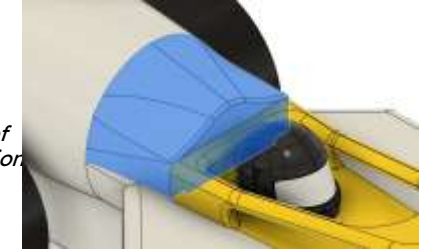


Figure below: Showing the use of the loft tool in fusion 360

Modelling Techniques

Autodesk Fusion offered a range of tools that helped us create a clean, compliant, and aesthetically pleasing car design. We primarily used sketches with extrude to form the main body, supported by dimensioning to meet regulations. Fillet and chamfer refined edges, while move and joint positioned and secured components. Loft was valuable for creating smooth transitions between faces, though revolve wasn't needed. Offset planes allowed sketches away from existing surfaces, and inspect ensured all measurements remained compliant. Throughout the process we repeatedly cut the car in half using the extrude function and then used the mirror tool to maintain symmetry.



Figure below: Showing zebra analysis from fusion 360

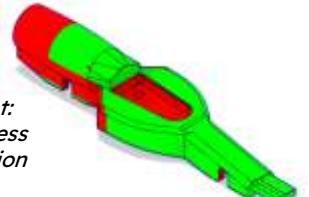


Figure to the right: Showing tool access analysis from fusion 360

CAD Organisation

As parts are manufactured at different stages, components were saved in separate files and imported into the main model, allowing them to be edited independently and efficiently. The joint tool positioned components accurately, and hiding parts made it easier to access obscured features. Most components were designed externally, but the front wing was created in the main file, then separated to ensure a clean, aerodynamic, and manufacturable transition.

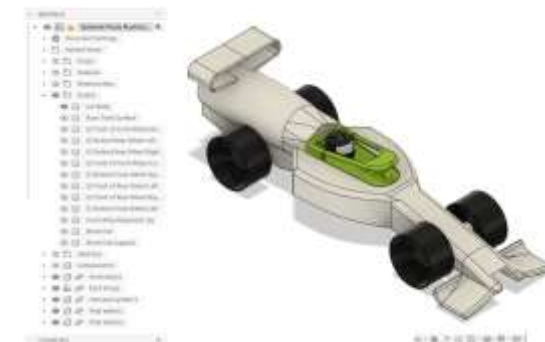


Figure to the left: Showing CAD organisation taken from fusion 360

All components and sketches were clearly named, and folders were used to keep the project organised. Fusion's timeline also helped us track and revisit changes throughout the design process.

Prototype 1



Figure below:
Showing a render of P1.

Key Features

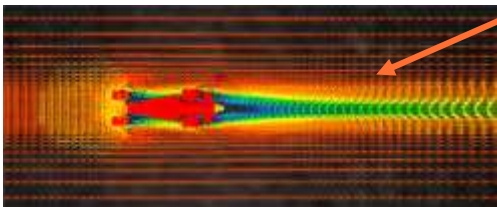
Our first prototype is our engineer Liam’s class car. In class, Liam’s car won with the fastest single lap with 1.1 seconds and won the overall race. His car also features a snow plow type front wing with the desire to guide air up and over the wings. Side pods designed to help guide air around the car with naturally curved surfaces. Teardrop inspired back wings were also incorporated into this car for it aerodynamic performance.

Liam ‘s class car was a successful model, with an excellent drag coefficient, and a good final mass. This provided an excellent prototype to base our design development on. We intended to lower the car resulting in an improved Cd as the body is lower to the track and has less frontal area. The car would not meet the competition regulations. These proposed changes will give us mass to spare, allowing for changes to front and back wings to be used, if needed.

Key Performance Data

CATEGORY	DATA	CHANGE
COMPLIANCE	FAIL	-
CENTRE OF MASS POSITION (mm) (from the origin x axis)	108.91	+6.14
TOTAL DRAG (Cdx)	0.293	+0.054
BODY (Cdx)	0.243	+0.048
CANISTER (Cdx)	0.167	+0.088
BODY AND WHEEL FORCE (N)	0.154	+0.008
FRONTAL AREA (m^2)	0.0032	+0.005
ESTIMATED RACE TIME (s)	1.153	+0.005
PROJECTED MASS (g) - (Before paint, glue, decal)	41.35	-4.43
RELIABILITY SIMULATION (FEA)	PASS	-

Figure below: Showing P1 in VWT



CFD Analysis

You can note a much narrower wash of disturbed air behind the car which strangely has resulted in an increased Cdx. This could be due to the stepper front wing angle.

Prototype 2



Figure below:
Showing a render of P2

Key Features

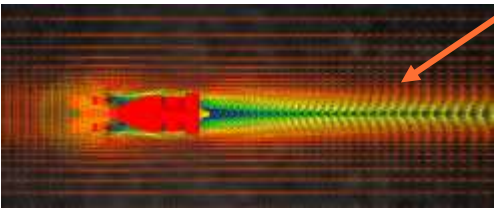
This evolved into the official car design, developed by Liam. This predominantly focused on creating a well performing aerodynamic, compliant car which was able to direct airflow through the car body, including the front, and rear wings. To elaborate, through the car’s front wing the air was able to be directed upwards, leading to the subsequent entrance into the hollow side pods and final travel through the rear wings, this produced a smooth airflow and subsequent aerodynamic wake.

Simultaneously, the car was able to produce a mass of 44.85gms and this allowed for significant modifications to be made whilst still meeting our mass target. Positive analysis including a low drag coefficient and an improved frontal area resulted in quicker racing times by 0.001s. This prototype was crucial for our success and featured improved data results.

Key Performance Data

CATEGORY	DATA	CHANGE
COMPLIANCE	FAIL	-
CENTRE OF MASS POSITION (mm) (from the origin x axis)	106.83	-2.08
TOTAL DRAG (Cdx)	0.233	-0.060
BODY (Cdx)	0.245	+0.002
CANISTER (Cdx)	0.107	+0.060
BODY AND WHEEL FORCE (N)	0.143	-0.011
FRONTAL AREA (m^2)	0.00319	-0.001
ESTIMATED RACE TIME (s)	1.152	-0.001
PROJECTED MASS (g) - (Before paint, glue, decal)	44.87	+3.52
RELIABILITY SIMULATION (FEA)	PASS	-

Figure below: Showing P2 in VWT



CFD Analysis

You can see here that the wash stream of disturbed air is narrower from P1 and the front wing angle is not a steep. Resulting in a better Cdx than P1 & P2.

Prototype 3 - State Finals Car



Figures below:
Showing a render of P3

Key Features

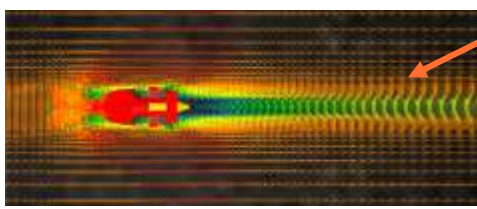
Our final car featured several changes. Initially, the front and rear wings we redesigned to produce a more aerodynamic car guiding the airflow around the car. Simultaneously, the car’s side pods included a hollowed floor producing a more lightweight car which directly influences the race time. Another major change was that the car’s body was lowered and it’s floor curved/filletted to allow air to flow without more Profile drag under the car.

This was supported by the performance data which displayed how the car performed continuously better in comparison to past car designs, hence proving that the changes made were beneficial for our team’s success

Key Performance Data

CATEGORY	DATA	CHANGE
COMPLIANCE	PASS	-
CENTRE OF MASS POSITION (mm) (from the origin x axis)	99.136	-7.69
DRAG (Cdx)	0.192	-0.041
BODY (Cdx)	0.199	-0.046
CANISTER (Cdx)	0.115	+0.008
BODY AND WHEEL FORCE (N)	0.118	+0.025
FRONTAL AREA (m^2)	0.00315	-0.00004
ESTIMATED RACE TIME (s)	1.145	-0.007
PROJECTED MASS (g) (Before paint, glue, decal)	42.25	-2.65
RELIABILITY SIMULATION (FEA)	PASS	-

Figure below: Showing P3 in VWT



CFD Analysis

You can see here that the wash stream of disturbed air is not narrower than the other 2 prototypes but the car still produced a faster predicted race time.

Final Car (VIJAY)

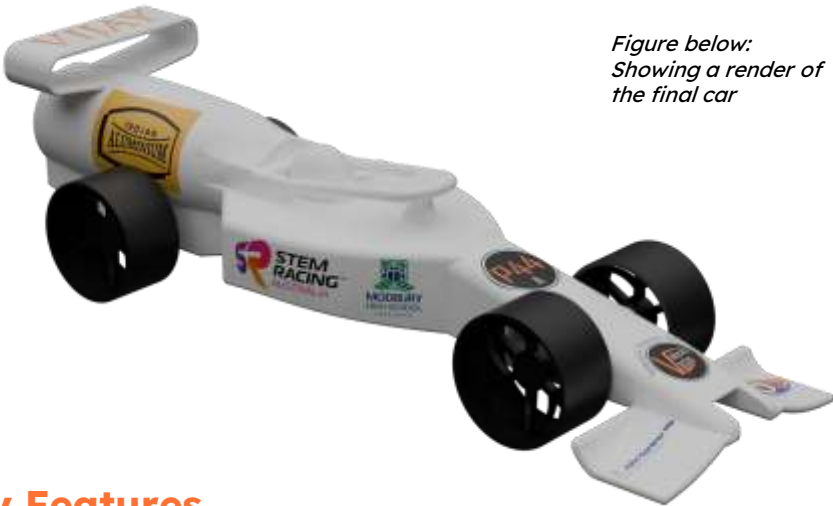


Figure below: Showing a render of the final car

Key Features

Our nationals car was designed to take aspects of the state car and improve upon them in every way. Some key features of the nationals car that has improved from the state car is the loft behind the halo. This loft was refined and improved from the state car as the lofting follows the outline of the halo. So as air passes the halo it is guided around the canister housing is follows a more generous path increasing laminar flow. Another key feature that has been improved upon is the front facing surface area has been decreased from our state car. High surface area results in more air needing to move around the car. The lower surface area will help to improve track time and reduce drag. We also reduced the overall width of the wheel base by 10mm again has helped us to reduce the frontal area.

Figures: Showing the difference in canister height, width of wheelbase and overall reduction in frontal area



Key Performance Data

Figure below: Showing data table of the final car

CATEGORY	DATA	CHANGE
COMPLIANCE	PASS	-
CENTRE OF MASS POSITION (mm) (from the origin x axis)	88.526	-10.61
DRAG (Cdx)	0.224	+0.032
BODY (Cdx)	0.128	-0.071
CANISTER (Cdx)	0.197	+0.082
BODY AND WHEEL FORCE (N)	0.138	+0.020
FRONTAL AREA (m^2)	0.00315	-0.00004
ESTIMATED RACE TIME (s)	1.150	-0.005
PROJECTED MASS (g) (Before paint, glue, decal)	37.25	-7.65
RELIABILITY SIMULATION (FEA)	PASS	-

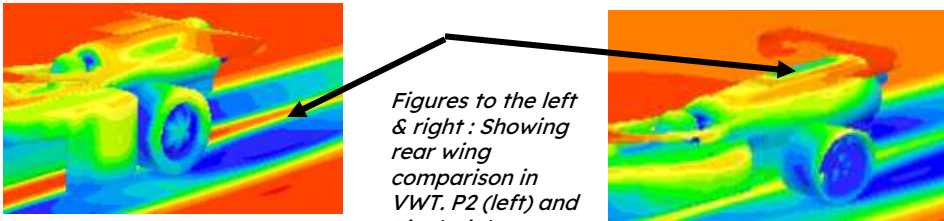
Rear Wing

The rear wing that we choose for our final car was an open airspace model, which we positioned on the top of the canister housing. We did this to keep the rear wing from further disturbing the airflow, meeting our mass target and being compliant.

CFD helped prove the concept, as can be see below, the airflow around the side of the canister is less affected by the positioning and design of our final wing.



Figure above: Rear wing from fusion 360



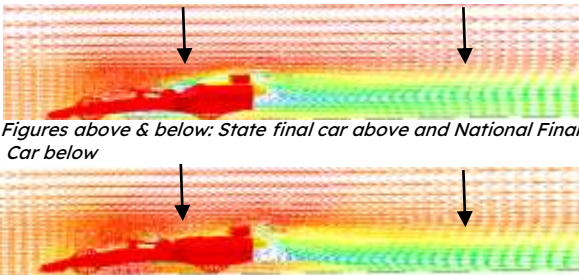
Figures to the left & right : Showing rear wing comparison in VWT, P2 (left) and Final (right)

Car Body

The final car body was designed to ensure that we maximised airflow, reduced frontal area and managed our mass. This was inline with engineering objectives 2 & 3.



- 1. Lofted canister housing to the back of the halo to ensure a more gradual path for the air to flow over.
- 2. Lofted progression from the nose of the car to the canister at the back to assist with laminar flow.
- 3. Arched under carriage to allow air to flow easier and reduce mass.
- 4. Filleted sidepods to assist with managing the mass target.
- 5. Tapered rear of the car with no pods to also better manage assist laminar flow.
- 6. Lowered the canister housing by 2.3 mm to help to reduce the frontal area of the car.

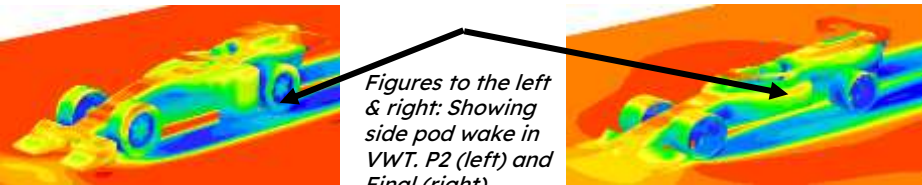


Figures above & below: State final car above and National Final Car below

In the images to the left you can see the benefits of the redesigned canister housing and loft to the halo. Creating less drag behind the driver and directing the flow back down after passing the canister. This reduces the wake of turbulent air.

Side Pods

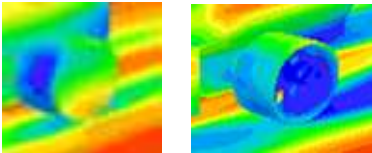
Our final side pod design balanced mass and aerodynamics to meet Engineering Objectives 2 and 3. Earlier open-cut prototypes reduced mass but directed airflow into the rear wheel. The final design enclosed this tunnel to reduce trapped air. Although this shifted mass rearward (Objective 6), underside fillets helped improve mass distribution.



Figures to the left & right: Showing side pod wake in VWT, P2 (left) and Final (right)

Wheels

The wheel design that we have opted for is a six spoked wheel with supporting structures to assist reliability and performance. This aligns with engineering objective 1 & 7.



Figures to the left & right: Showing wheels in VWT, P3 (left) and Final (middle). Image 61: Final wheel prototype from fusion (right)



Whilst you can clearly see that the enclosed wheel caps were more aerodynamic we chose the spoked wheels to assist with our mass target and to help to maximise the launch phase of the car.

Axle System

We chose a stub axle design which is glued into the body using alignment jigs. This allowed us to clear out the the undercarriage of the car to provide better laminar flow. This to aligns with engineering objectives 3 & 7.

Figures to the right: showing our axle systems



Bearings

We tested three types of bearings and ultimately chose hybrid ceramic bearings from Boca, as they performed best during testing our engineering objective 1.

Figure to the left: Tether guide

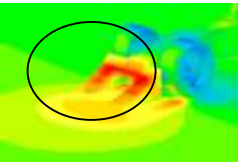


Tether Guides

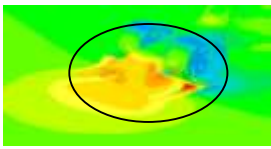
Our tether guide choice was made based on reliability and performance. We wanted a basic and known system so that it was easy to align and still provide good performance addressing engineering objective 5, 6 & 7.

Front Wing

The front wing was designed to match the car's minimalistic style while providing wind shielding and reliability. Testing showed that overly pronounced designs increased Cdx, so a balanced approach was used to meet Engineering Objectives 3 and 7.



Figures to the left & right : Showing front wings in VWT, P2 (left) and Final (right)



Overall

All decisions made about the final car design have been done so with engineering objective 4 in mind. This has been the overarching objective in the engineering process.

Evaluation Against Engineering Objectives

- 1. Having accurate wheel and axle systems ✓
- 2. To have a mass of 50.5g total mass on our final car ✓
- 3. In the software CFD to have a drag coefficient lower than .2
- 4. To have met all race regulations and specifications ✓
- 5. To have a reliable and working tether guiding system ✓
- 6. To have a reward centre of mass on the car ✓
- 7. To have on track reliability and safety when the car goes down the track ✓

Digital Analysis / CAD Inspect Tool

- We used the Inspect tool to get the following analysis data:
- **Centre of Mass** - This is an important feature for car design. To minimise the tipping moment on our car when on launch, we needed to locate the CoM towards the rearward of the car, in all three axis. The closer we can get the CoM to the line of thrust, the more efficient the launch sequence is.
 - **Measure** - We Measure to check the accuracy of our geometry during the design process. The checking of these dimensions were also critical for our important compliance checks.
 - **Draft**- This helped us look at the manufacturability of our car during the CAM process.
 - **Accessibility tool** - We used this to check that all parts of the car design will be able to be accessed during machining.
 - **Minimum Radius Tool**- Check geometry re < 3mm radius for problems with a 6mm Ball Mill

Finite Element Analysis

Finite Element Analysis (FEA) is a method to estimate or predict how products are likely to perform under stresses encountered during their intended use. We used FEA on our rear wing, as this may encounter stresses during the deceleration process. We used Fusion to conduct the FEA simulation

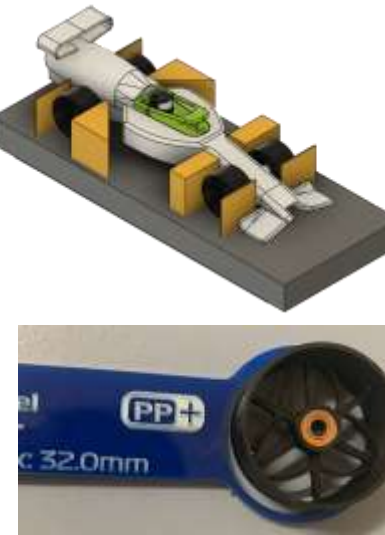


The images to the left are some of the FEA test we conducted on or parts. We applied 8 newtons of force in our testing. We know after using the F=ma (Force = mass (50.5 grams) times by acceleration (19m/s) results in 0.96 Newtons). We decided to increase the force encountered by a factor of 8 to be confident that are parts can withstand racing on the track

Figures to the left: Showing components in FEA results

Compliance Checks

As compliance with the Technical Regulations is a vital part of this competition, we analysed our designs both in Fusion and after we made the prototype cars on the Mill. We used digital verniers, 2mm and 3mm drills, some laser cut jigs and a digital height gauge to help. Using Fusion, the Drawing workspace helped us check compliance and accuracy easily. We had trouble with the height of the car, so the Halo and Front wing were both out, and would have caused a massive points loss and time penalty if not modified.



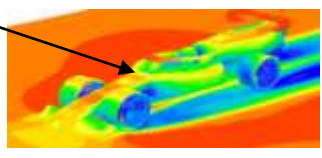
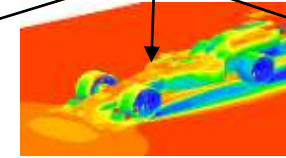
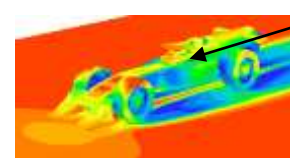
Figures below and to the right: Showing compliance checks both physical with jigs and digital in fusion 360

Computational Fluid Dynamics

CFD is an important design and analysis tool, as poor aerodynamic performance is a loss. To conduct our testing we used Virtual Wind Tunnel as we experienced issues with our other platforms. VWT was helpful in producing excellent data and images of the drag a force acting on the car and components. This was heavily relied upon to guide the engineering process. Some of these are:

Body Surface Velocity - Profile Drag

This indicates the wind speed as it approaches the car and then how it interacts as it passes over and through surfaces. The wind speed is set at 18 m/s which is indicated as orange (the same colour before it hits the car). What we want to see is a consistent orange colour over the top surfaces of the car meaning that the wind has minimal disruption as it passes the car.



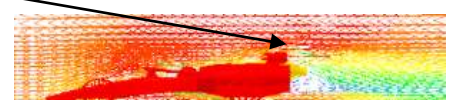
Figures above: Images from VWT

P1 National Car P2 State car

This data helped us to shape the front loft of the car and ensure that the front complimented the final design.

Wind Velocity Y Plane

The wind speed on the Y Plane of our model allowed us to look at areas where the wind was getting trapped and therefore aerodynamic losses. We looked at the rear of the car in particular. Here we are looking for the wind to return to the speed it entered the car as quickly as possible. In the images below you can see that there is a significant area of blue (disturbed air flow) behind P1 compared with the final car.



Figures above : Images from VWT

P1 Final Car

This data helped us in the design of the tapered undercarriage at the rear of the car. This made a significant impact on the over reduction in the drag of the final car.

Wind Velocity Z Plane

Much like the wind speed on the Y Plane we used these images and data to fine tune the air flow around the front wing and sidepods of the car. The theory is the same as on the Y plane, with the aim to return the air to the original speed and direction it was on before it interacts with the car as soon as possible



Figures above : Images from VWT

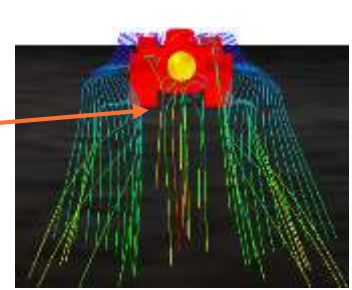
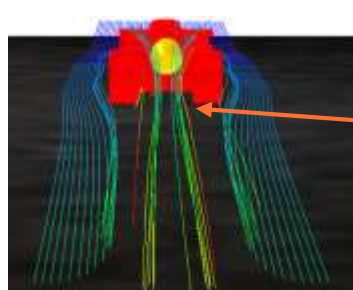
P1 SF Final Car

The three areas of focus for us here were behind the front wheels, behind the side pod and at the rear of the car. As you can see there is far less blue 'disturbed air' in our final car compared to P1. Again, significantly helping the overall drag.

Streamline Imaging

When analysing streamline data from a car wind tunnel test (real or virtual), you're essentially looking for how the air *moves around and interacts with* the vehicle. Good streamline analysis helps you understand drag, lift, flow attachment, separation, and turbulence, which all influence performance.

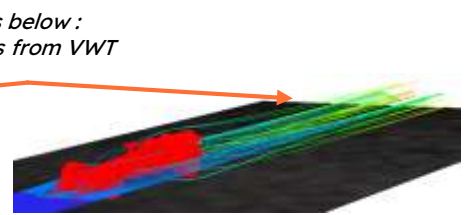
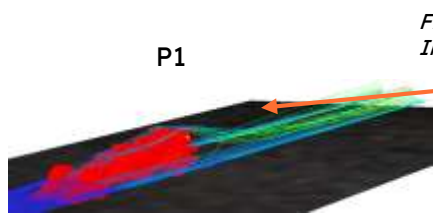
In the images below we can see firstly that P1 has even flow symmetry meaning there are few geometrical flaws, whereas P2 has a few inconsistencies. The below images also highlight has a smaller wake, with less turbulent airflow (tangled streamline).



Figures to the left & right : Images from VWT

P1 P2

Furthermore the streamline images help to highlight the narrower wake of P1 compared with P2. This would be due to multiple factors with a couple of these being the tapered rear of the car on P1 and different rear wings. This is reflected in the overall Cdx in the graph below.



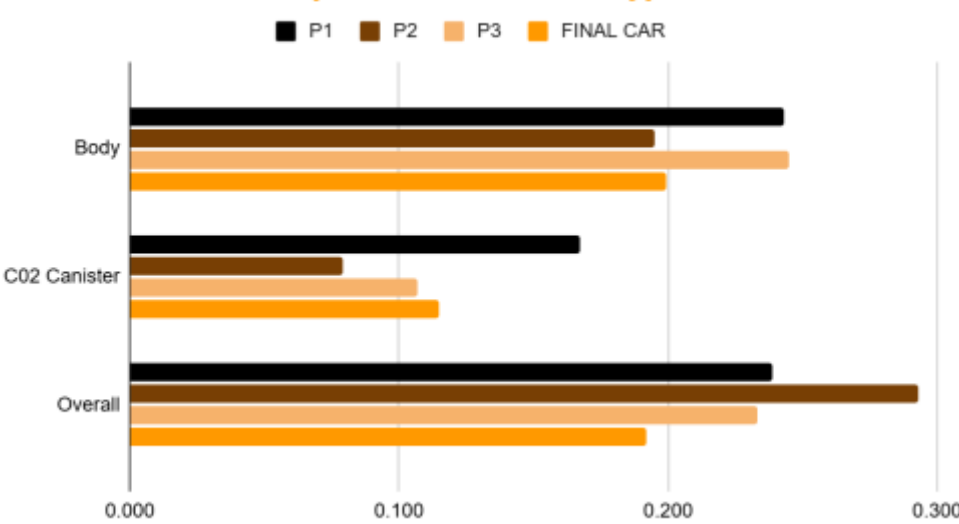
Figures below : Images from VWT

P1

	P1	P2	P3	FINAL CAR
Body	0.243	0.195	0.245	0.199
C02 Canister	0.167	0.079	0.107	0.115
Overall	0.239	0.293	0.233	0.192

Figures to the left & below; Data from VWT

Component Cdx on Prototypes



Manufacturing Goals & Objectives

- 1. Producing three identical cars.
- 2. Each car weight no more than 51 grams.
- 3. Limiting having to recut or reprint aspects of the car after beginning manufacturing.
- 4. Gain maximum points for scrutineering.

Machines

When we progressed to manufacturing our car, we used the Denford router 2600 pro. We used this machine to cut the main body of our prototypes and final cars. This involved testing and resetting the offsets on the machine to ensure accurate machining. Other machines we used include the Bambu Lab X-1 carbon and the Bmbu A1 3D printers. The use of the original prusa SL1S speed resin printer was used for the front wing. We used these machines to print prototype and final parts for the cars which include the front and back wings, grommets and wheels.

Figure to the right: Picture of the offset positions used on our machine from VR Milling.



CAD based design software

While designing our car we used a software called Autodesk Fusion 360. We chose this software for it's vast tools and accessibility. While in our first year of the class we created our class cars using Fusion and all members of the group were familiar with the software and felt it was a safe reliable choice.

Coding the machine

Using an app called QUICKcam Pro to run the 3D design we created and write code for the machine to use. After setting the orientation, depth of cuts and what type of cut, we then post processed to get code to cut the block. Each profile of the car is post processed to be cut, using separate codes. We name these code files, starting at 0001 and ending in 0004. Each individual side of the car's code is then edited to tell the machine to cut axle holes in specific location, mirror sets of code and looks for the next set of code.

```
G00 A0
M98 P0001
G00 A90
M98 P0002
G00 A180
M98 P0003
G00 A270
M98 P0004
G00 A0
M30
```

Figure above: Sub Call from VR Milling

Paints and primer

Tamiya paint is an acrylic paint, one of the three types of paint that we were choosing from. They were acrylic, enamel and oil based and there isn't a lot of difference between the paint's besides some being stronger than others and some lasting longer than others and different finishes. We are using tamiya paint, which from previous teams experience at school was of high quality in terms of finish and ease of application.



Figures below :Showing the paint used and finished painted prototype

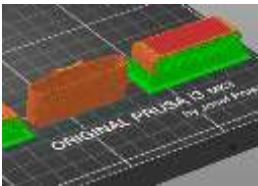


3D printing

3D printing is a crucial part of the manufacturing of the final car. Without 3D printing the car would be missing crucial parts including the front wing, back wing, wheels, grommets and halo. For our 3D printing we used a combination of the Bambu Lab X1 carbon, Bambu Lab P1S and the Bambu lab A1 printers for testing our designs. The front wing (middle image) has been printed using the resin printer. We also used a variety of materials and printed in a variety of orientations and infill levels to achieve the best print.



Figures above: 3D printed parts and slicing



CAD/CAM Manufacture:

We learnt how to write Computer Numeric Code (CNC) during our school lessons.

We know that the first job is to ensure that the geometry of the car is the correct size, to fit into the Balsa wood blocks. This includes overall length, width and height. Our blocks are 225mm x 65mm x 50mm in size.

We studied feedrate, spindle speed and depth of cut as these terms relate to machining

Feed Rate:	2250mm/m	default
Spindle Speed:	23000rpm	default
Depth of Cut:	5mm Max	
Step Over:	.2mm	default is .9mm



Figure to the left: Showing the difference in machine a model at .9 stepover (front) compared with .2 (rear)

We use the software program QuickCAM Pro to write our code, and have the use of a Denford 2600 Router. This machine enables us to rotate the block to access, top, bottom and side machining without having to turn the block by hand. We use a series of codes, joined together called a sub program, to rotate the work on the chuck.

We used a 6mm diameter ball mill for all of our machining. The sub program was written for us, as all students use it in our class. This is the image illustrating the tool paths and final appearance of our car, from the QuickCAM software.



Figure below: Showing machining errors and the Machine plot from Quick Cam



Machine Tolerancing

We used the IT14 internal tolerancing grade to give us an idea where to start. Machine tolerance is the maximum and minimum variation a component can be from the original nominated exact size. This means that as long as the component or part, after manufacture, is within these limits, it will be an acceptable and workable part.

We have applied the following tolerances:

CNC machining:	+/- 0.1mm
3D Printing	+/- 0.1mm

Quality Control

Block Selection: Even though we had a limited number of blocks available, we chose the selected blocks which accurately prepared, the correct mass and free of any knot defects.

Compliance Scrutineering: The geometry was scrutineered in CAD form, then physically after machining and then after assembly. This was done by our engineering team.

Additional scrutineering: Another member of the team was given the job to also scrutineer the cars.

Quality of finishing and assembly were closely looked at.

Car Manufacturing Process

We followed the processes we used with our class cars. However there was a lot more work went into designing and developing our national final car. The most important change was the need for the car to be compliant with the STEM Racing technical rules, so we tried to check these key measurements throughout the process.

Safety was always on our minds.

Our Manufacturing process was:

1. Design the car and check compliance using CAD. Ask our teacher for help if required.
2. Finish the geometry and write the code
3. Make a prototype car, to check that the machine is working accurately.
4. We have learnt that the manufactured car is sometimes different than the drawings, so it was important we measured the car carefully after machining.
5. The machine was working accurately.
6. We then fitted the Front Wing, and found we had an issue with 3mm clearance/airspace and had to make changes. We changed the position of the axle holes to help. We used PVA adhesive.
7. Next we fitted the Halo, and again we needed to adjust the axle position to make the make the Halo the right distance from the track.
8. The Rear wing was fitted next checking rule compliance making it looked straight.

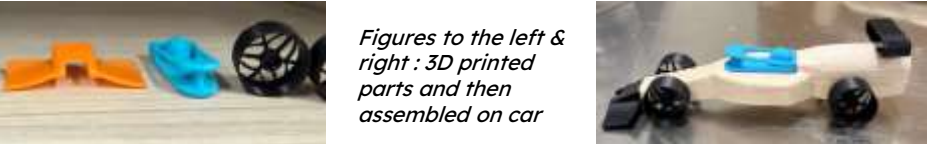
The wheel and axle system required careful fitting of the bearings and the wheels themselves. The tether guides were glued in place. We sanded our car using fine grade abrasive paper, before applying undercoats and top coats. We weighed the car several times to check if we were meeting our mass target.

Finishing Process

As part of the final manufacturing process, after the assembly of the front and rear wings, we began preparing our car for painting. As we used a finishing machining plan (canned cycle), there were few marks to remove from car body. However we needed to make sure that we kept the machined shape and that we didn't remove too much wood and make the car too light. We used 240 grit Aluminium Oxide abrasive paper to begin the process, which is a medium grade abrasive to remove any machining marks remaining. This was followed by 400 grit for a final smooth finish. This was followed by sprayed coats of high quality paint, weighing the cars between each coat.

Assembly

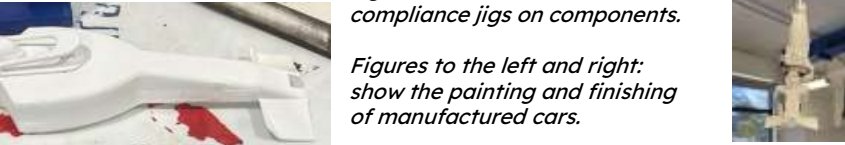
We tested our car assemblies using custom-made jigs, along with precise marking and measuring tools such as digital verniers and machine blocks from the Design and Technology department. We 3D printed our parts and glue the relevant pieces in place.



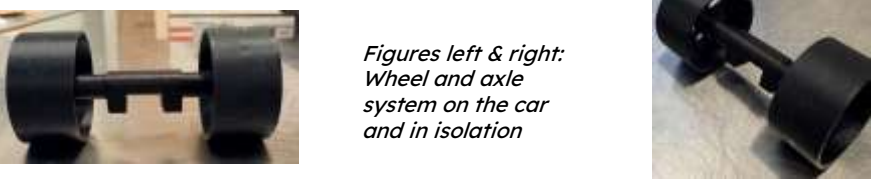
Once parts were glued and set, the use of compliance check jigs we put into use. Each jig corresponding to a specific regulation, create a sort of checklist to go through after and during each cars assembly. The jigs were laser cut out of acrylic in our schools woodworking building. After each car passes the jig compliance check it moves onto the next stage of painting.



Our primer and paint process consisted a methodical process in which we recorded the mass of the car after each action, where that was painting or after sanding. You can see this on the mass calculation spreadsheet to the right. This process allowed us to monitor our progress towards our target mass.



The next step in the process is to again weigh the cars fully assembled to ensure that not too much mass has been lost due to evaporation and then to apply the stickers. We then apply a couple of coats of clear or clear pearl paint to put a final layer over the stickers to help with skin friction drag and improve the aesthetics



After the painting process is complete and the target mass has been achieved it is time to assemble the wheel and axle system. This is a very important aspect of the process. As we have previously stated in our engineering philosophy, we believe that having an accurate wheel and axle system is the most important part of ensuring we have a fast a reliable car. Once assembly is complete a final weight check is done and then another in 2 days to ensure that the mass of the cars is relatively stable. This then completes the assembly process. For more information on the process please see page 9. We then were able to track test the car (prototype). Please see the results in the right hand side column.



Figure left: Car on track during testing running against our state car

Physical Testing
Mass Calculations

As car mass is one of the most critical performance factors, we used CAD to assign physical materials to every component and adjust their densities where needed. This allowed us to generate an accurate ballpark estimate of the final assembled mass. For instance, this process helped us determine whether using 12g brass rear axle tubes would push the car over the 50 g limit. We also used the predictive section of our mass-calculation spreadsheet (shown below) to select the appropriate block mass. During manufacture, we regularly weighed each component to maintain strict quality control and ensure our predicted mass matched the real car.

VELOCITY VORTEX MASS CALCULATIONS CAR A									
PROJECTED FINAL CAR MASS CALCULATIONS				PROJECTED CAR BODY MASS CALCULATIONS				MASS PROGRESSION	
Original Block Mass	65	Block Mass (BM)	65	Off the machine	9.225	30.395			
Front Wing	4.00	Projected Block Mass (AM)	9.225	After Seal and Assembly	34.765	14.855			
Axles	2.00	Mass % Machined	85.30%	After Primer 1	32.34	7.260			
Bearings	7.00	Block Volume	7.025	After Sand	31.95	7.670			
Wheels	8.00	Car Volume mm3	1.025	After Primer 2	35.7	4.430			
Halo & Helmet	5.12	Volume % Machined	85.20%	After Sand	34.93	4.600			
Car Body	9.225	Block Density	89.730	After Paint 1	36.5	3.130			
Rear Wing	4.20			After Sand	36.3	3.430			
Tether Guides	0.60			After Paint 2	37.55	3.070			
Greenstuffs	3.20			Target Mass					
Paint/Stickers/Glue (Allowance)	12.80								
Car Mass (No paint)	37.650								
Target Mass	50.5								
Difference	12.850								
Car Body Mass before final Assembly	43.850								

Figure above: Mass calculation spreadsheet

Block Mass		3D Printing Parameters	
110	80	200% infill (15% left)	Begin
Front Wing		9.40	4.39
Axles		2.44	
Wheel and bearing		10.08	
Halo & Helmet		5.3	
Car Body	10.37	9.58	9.89
Rear Wing		4.00	3.99
Tether Guides			1
Paint/Stickers/Glue (Allowance)			1

Figure left: We designed a spreadsheet to help find the best configuration of components to give us the ideal mass before painting.

Block Accuracy

Before manufacturing begins, blocks are measured using a digital caliper to check the canister hole is centred. The block is then checked against regulations (65 mm wide, 50 mm deep) and weighed to confirm mass compliance.

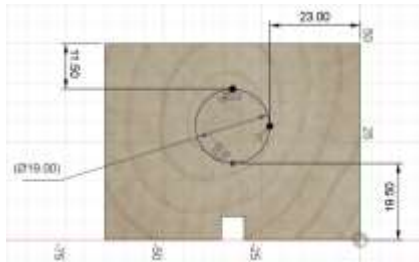


Figure above: Physical block accuracy test

Machining Accuracy

After cutting cars for either prototypes or final car we measure the physical model against the CAD one to ensure that we has the machine offsets correctly set and to ensure that we still had a compliant car.

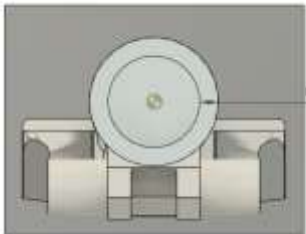


Figure to the left & right Physical machine accuracy test



Canister

Although the canisters appear consistent, their weights can vary by up to 1 g. When examining used canisters, we also found that the puncture holes were not always the same size or perfectly centred. To minimise the impact of these inconsistencies, we used the starting mechanism and launch jigs to ensure a controlled and repeatable setup. We also discovered that the CO₂ inside the canisters is stored under high pressure and exists as a liquid before rapidly turning into gas when released.



Figure to the left: Physical canister hole accuracy check

Figure below: Car misaligned on track

Starting Mech Aligned at Launch

Launching the car perfectly straight is crucial. If the car begins at an angle, it will try to turn, increasing contact with the tether line and creating additional drag, which slows the car down.



Car Aligned at Launch

We used a spacing block to ensure the cars were perfectly parallel to the track for every launch. Consistent, accurate alignment with the starting mechanism firing pin is essential for reliable engagement and fair, repeatable race conditions.

Tether Guide Test

To compare tether guide materials, we slid a brass eyelet and an alconite ceramic ring down a fishing line; the faster-moving one indicated lower friction and therefore less drag during racing. We also inspected each internal surface under magnification to assess smoothness. Although the ceramic ring was faster, we prioritised ease of assembly and went with the brass eyelets.

Track Testing

We were able to test the prototype of our final car on tack and were happy with the performance and the reliability of the car as a whole and individual components. We also know that there is more performance to be found from the improved final bearings.

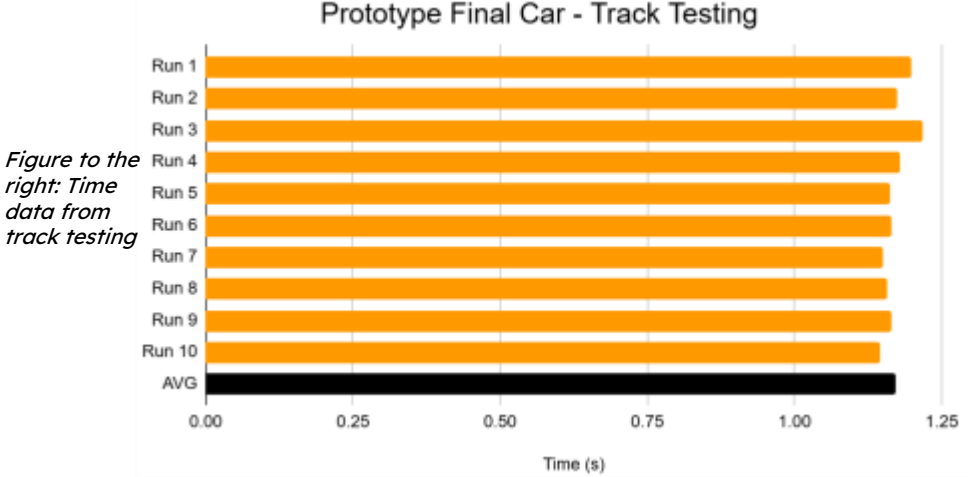


Figure to the right: Time data from track testing

Requirements Analysis

We identified what are the losses in STEM Racing cars, and based our research and development on minimising those effects. We summarised them as Mass, friction aligned to the tether guide system, track contact, aerodynamic influences and wheel and axle systems. In addition, compliance with the regulations was critical. Ease of manufacture and assembly were also considered. As we are studying Physics, the basic concepts and rules were quite easy to understand and apply.

Ideas

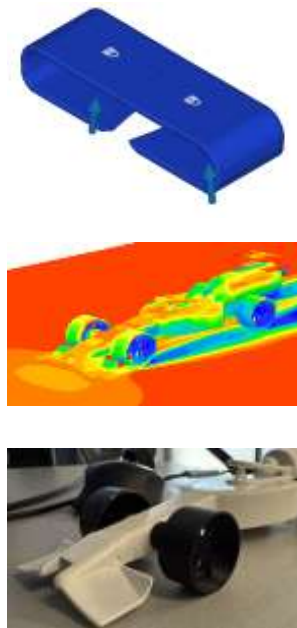
The Car is made of several components which we developed through continuous testing using CAD. To initiate, the body was aimed to be a thin, lightweight and smooth to allow for an advantageous airflow around the car. Accompanying the body, the wings were mainly designed around compliance and mass, and we needed them to be quite simple in concept, as we saw no aerodynamic advantage in complex designs. The wheel and axle system ended up as a stub axle design, a change from the State final, we wanted to have the wheel rotate on a small axle, saving mass. We didn't change the tether guide system, which worked well at the States.

Development

The final development of the car and its components was developed through continuous testing and remanufacturing of the wings and body. We had several setbacks which delayed our progress including the mass of our wings. Throughout CAD was used effectively to check compliance and mass.

Analysis

Analysis was a key tool in allowing us to develop and change each major component of the car. For example, through CAD, combined with several applications such as CFD and FEA, we were able to identify possible weak points on our wings, and further analysis was conducted by physically testing of the prototypes. In CFD, we mainly used Cd results, but looked at Frontal Area, lift and downforce data Accompanying the analysis of the wings, major analysis on the car's body was done through FEA, where we applied a force of 8N to conduct the tests. CAD helped with mass and compliance checks throughout, and we found this to be accurate and helpful. Overall, we were able to provide some detailed analysis on major components of the vehicle.



Figures above: Show the FEA of rear wing (top), Car in VWT (middle) and Car finished as prototype (bottom) highlighting our varied analysis

Body

We didn't change the final car body much from the States, except the area immediately behind the Halo to improve Aerodynamics and remove mass, however some minor changes were made to suit regulations. Some of the changes we did make included changing the wheel location on the car, and changing the size of the wheel's bearing holes to suit our new wheel and axle system, with our purchased bearings. Most of these changes were crucial to create a car to satisfy us and to meet the competition's regulations. Overall, we are happy with the final body of the car.

Physical Testing

We conducted a variety of physical testing on our prototypes and final versions of our car. These tests include:

- **Tether Guide** - The tether guides were test to experiment with different materials, but we found that our State brass eyelets worked well, and there was no need to change the positioning of them on the car body to prevent yaw.
- **Block Accuracy** - We successfully tested our block accuracy as the canister hole is sometimes not concentric, affecting our car manufacture after rotation. This was done to eliminate machining errors of the car body and to stop breaches of compliance, especially the 3mm around the canister housing.
- **Machining Accuracy** - It was very helpful to have a digital vernier to check the accuracy of the cars coming off the mill, as this could affect our car's compliance and assembly.
- **Track Testing** - track testing was a valuable form of evaluation particularly for the tipping moment, tracking and reliability of front and rear wings upon deceleration. We don't have a halo deceleration device, so we had to use the brushes.
- **Checking Compliance** - checking compliance of the car this time around was a lot easier with the introduction of our jig's. Often any issues with compliance of the car was picked up prior to physical testing but having the jigs gave us piece of mind that our car is compliant with stem racing regulations.

Manufacturing and Machining Process

We machined our cars using a 4 rotation machining process with a 6mm Ball nose tool. We used a 0.2 step over which required a light sanding post machining. We had an issue, to do with the axle hole positions and car length, affecting the depth of the canister housing. However, we found QuickCAM Pro and Virtual Milling software easy to use, including the programming of the canned cycles for block rotation.

Front and rear wings were made at school using our 3D printing lab. This process went smoothly with some redesigning and reprinting of parts required to align with regulations standards. The parts were accurately printed.

Wheel and axle assembly - Wheels and axles were made at school using our 3D printers, one hiccup with our wheels was printing older versions of our final wheel which were made to fit a smaller bearing. One issue we ran into while printing axles was deformation of the bearing housing, meaning the attachment point wasn't fully round. We fixed this by changing the printing orientation having the axles positioned upright.

Assembly of cars - This was greatly assisted by a genuine understanding of or target result for our cars and the use of assembly jigs. Both cars passed scrutineering following assembly.

Finishing - This process was closely aligned with our target mass goal. We are happy with the final build and appearance of our cars.



Figure to the right: Show machining errors whilst writing the NC codes

Tolerance & Quality Control

Tolerance and quality control was an attitude towards high standards throughout this process. Being aware of tolerancing and allowance in design and manufacture were necessary for us to produce our two race cars that are compliant, and have met our mass target. We applied a +/- 1mm tolerance on components which had been 3D printed or created through the CNC machine, this helped us manage errors while remaining within regulations. For example, we added 4mm thickness to our chamber housing which allowed it to remain within regulation while also adding on extra support to the chamber housing.

We used jigs where possible during assembly to ensure accuracy and reliability. This worked well allowing us to produce both cars easily and successfully. We ensured that the quality of each component was well managed throughout our journey. Liam, our lead engineer ensured that the wheels and bearings were compatible prior to manufacture or adding them onto the wheels. Hence, quality control was a technique which was beneficial for the development of all components.

CAD

Liam was able to undergo a process which elevated each aspect of his CAD skills. To elaborate, he was able to extend his skills from those which he gained while undertaking the school subject. This was possible due to the continuous support and advice which he received from key mentors who taught him of the several techniques on how to create a well analysed car. Overall, through the journey, Liam was able to successfully lead the engineering department with a clear understanding of each topic.

Compliance

To strengthen our interpretation of the regulations, and ensure we were compliant with our car, we engaged in discussions with Mr. Stephen Read, an experienced STEM Racing mentor and educator. Drawing on his extensive experience, he provided valuable guidance that supported Liam, our Lead Engineer, in reviewing and refining the car design to ensure it aligned with competition regulations.

Rendering

Rendering was not a major difficulty however, however minute issues were faced during the process. For example, Sebastian initiated the rendering process through Fusion, however, due to errors with the decals, he transitioned to Blender which offered higher quality renders with detailed decal position. This allowed us to develop high quality renders with accurate sticker location.

However, due to the render being developed in Blender, it was not possible to transfer the file with decals to Fusion and hence the renders in the compliance folio were simply a photo of the final car. Overall, rendering was simple with minimal issues.



Figures to the left: Final Car Render developed from the Blender Application.

